

Homework 1

2.1.1.a Let μ be a positive inner regular locally finite measure. Suppose there exists a constant D such that for all $x \in \mathbb{R}^n$ and $r > 0$ we have

$$\mu(3B(x, r)) \leq D\mu(B(x, r)).$$

Define the weak $L^{1,\infty}$ norm $\|\cdot\|_{L^{1,\infty}}$ by

$$\|f\|_{L^{1,\infty}} = \sup_{\lambda>0} \lambda\mu(|f| > \lambda).$$

We will show the uncentered maximal function M is continuous $L^1(\mathbb{R}^n, \mu) \rightarrow L^{1,\infty}(\mathbb{R}^n, \mu)$. Define $E_\alpha = \{x \in \mathbb{R}^n : Mf(x) > \alpha\}$. We need to show

$$\|Mf\|_{L^{1,\infty}} \leq C\|f\|_{L^1},$$

and it will suffice to show for some $\alpha > 0$ that

$$\alpha\mu(E_\alpha) \leq C\|f\|_{L^1}.$$

First, I claim E_α is open. For $x \in E_\alpha$, there must be a ball B_x containing x such that the average value of f on B_x is greater than α . Then this ball is in fact contained in E_α , since for any point $y \in B_x$ we can choose a ball $B_y \subset B_x$ containing y , and this proves $Mf(y) > \alpha$.

Now use inner regularity. Choose any compact $K \subset E_\alpha$ and cover K in balls $\{B_x\}_{x \in K}$ such the average value of f on every B_x is greater than α . Choose a finite subcover $\{B_i\}_1^N$ and choose a further subcover $\{B_{i_k}\}_1^M$ such that $\cup B_i \subset \cup 3B_{i_k}$. Then

$$\mu(K) \leq \mu\left(\bigcup_{k=1}^M 3B_{i_k}\right) \leq D \sum \mu(B_{i_k}) \leq \frac{D}{\alpha} \sum \int_{B_{i_k}} |f(y)| dy \leq \frac{D}{\alpha} \|f\|_{L^1}.$$